Wenlong Xiong

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Skills

Languages: Golang, Python, Java, Bash

Technologies: Kubernetes, Kafka, gRPC, Envoy, Terraform, SaltStack, Airflow, Spark, Hadoop, Presto/Trino

Experience

Robinhood, Senior Software Engineer

Service Discovery and Traffic Feb 2023 - Present

- Implemented an egress proxy service for egress filtering, and leading migration effort to secure outbound traffic across 100+ services and multiple protocols (https, ssh, ftp, websocket).
- Enhanced existing kubernetes controllers to generate Calico network policies for egress controls.
- Maintained and supported gRPC Golang / Python libraries, gRPCLB / XDS lookaside loadbalancers.
- Designing and implementing a websocket gateway to stream backend events to Web clients.

Kafka and Streaming Platform Jun 2021 - Feb 2023

- Helped design and implement RH's Golang Kafka library, with features like deadletter queues, observability by default, and multicluster produce and consume.
- Drove RH's multicluster kafka initiative and migrated all critical trading services to multicluster produce / consume, removing a kafka cluster outage as a possible single point of failure for trading.
- Designed and implemented a Kafka consumer proxy sidecar in Java, which increased consumer stability and reduced rebalance delays on the trading path, while matching existing throughput / latency.
- Directed Build vs Buy decision after evaluating vendors (Confluent Cloud, MSK) & open source (Strimzi, etc). Helped rightsize existing Kafka infrastructure, saving 2.5MM/Y. Defined migration path to running Kafka on kubernetes, which is serving X% production traffic.
- Worked with a small team to authn/z all kafka traffic, migrating ∼100 services across a dozens of clusters.
- Helped implement a CDC system using Debezium & Flink for event-driven applications.

Data Infrastructure Feb 2020 - Jun 2021

- Oversaw RH's Spark/Hadoop multicluster (1 to 15) migration, and built initial version of resource based job routing system. This enabled RH's first data-intensive product (IPO Access), and met compliance requirements for IPO (financial data access controls).
- Developed a dependency ordering framework for managing the lifecycle of Kubernetes sidecars, now used in all RH production kubernetes services (100s of namespaces, dozens of clusters).
- Bootstrapped Robinhood's Data Engineering team & interviewed first ∼10 initial hires.
- Drove Airflow 1.x to 1.10 upgrade and kubernetes migration for hundreds of DAGs, dozens of services.
- Derisked and started kubernetes migration for production Presto/Trino clusters.

Microsoft (Azure Stack), Software Engineering Intern Jun 2019 - Sep 2019

- Compute infrastructure (VM provisioning) team for hybrid cloud. Enabled IMDS (Instance Metadata Service) in Azure Stack VMs, and extended existing Azure Stack .NET services to support this newly introduced component.

Bloomberg, Software Engineering Intern Jun 2018 - Sep 2018

- Wrote an event ingestion service in Python that performed real time filtering and redaction on messages sent using Instant Bloomberg Chat.

Links & Talks

Current Summit, Kafkaproxy: Decoupling Kafka Consumer Logic from Application Business Logic 2023

Current Summit, Dead Letter Queues for Kafka Consumers 2022

Education

University of California, Los Angeles

M.S. Computer Science; GPA: 3.93 Sep 2018 - Dec 2019

B.S. Computer Science; GPA: 3.76 (*Cum Laude*) Sep 2014 - Jun 2018